Cahit Ikbal Er

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Professional Summary

Ph.D. candidate in Mechanical Engineering working on planning and control for heterogeneous multi-robot teams. I design scalable algorithms with provable feasibility, robustness, and complexity guarantees for energy-constrained, long-horizon UAV-UGV missions such as aerial monitoring, coverage, and collaborative exploration, and validate them in simulation and on real hardware.

Education

Northeastern University — *Boston, MA*
PH.D. IN MECHANICAL ENGINEERING *(Concentration: Mechatronics)* — *Expected: May 2028*

Northeastern University — *Boston, MA*
M.SC. IN ELECTRICAL AND COMPUTER ENGINEERING *(Concentration: Robotics)* — *May 2026*

Bilkent University — *Ankara, Turkey*
B.SC. IN MECHANICAL ENGINEERING — *June 2023*

Publications

- [1] **C. I. Er**, A. Kashiri, Y. Yazicioglu. **RSPECT: Robust and Scalable Planner for Energy-Aware Coordination of UAV–UGV Teams in Aerial Monitoring.** J. Intell. Robot. Syst. (**JINT**), 2026. [paper] [video]
- [2] **C. I. Er**, S. Juttu, Y. Yazicioglu. **Energy-Aware Collaborative Exploration for a UAV–UGV Team.** International Conference on Intelligent Robots and Systems (**IROS 2026**). [preprint] [video]
- [3] R. Fowler*, **C. I. Er***, B. Johnsenberg, Y. Yazicioglu. **PRO-SPECT: Probabilistically Safe Scalable Planning for Energy-Aware Coordinated UAV–UGV Teams in Stochastic Environments.** International Conference on Intelligent Robots and Systems (**IROS 2026**). *Equal contribution. [preprint] [video]
- [4] **C. I. Er**, Y. Yazicioglu. **Robust and Scalable UAV–UGV Planning for Aerial Monitoring under Energy and Precedence Constraints.** American Control Conference (**ACC**), under review. [preprint]

Experience

Multi-Agent Robotics and Autonomy Lab (MuRAL), Northeastern University — *Boston, MA*
RESEARCH ASSISTANT — *Sep. 2023 – present*

- Designed RSPECT [1], a polynomial-time UAV-UGV planner with feasibility and robustness guarantees: 100 points, 10 teams in 3.4 s; beats tuned simulated annealing and genetic algorithms by up to 24%/62% at ~1000x less compute.
- Co-first-authored PRO-SPECT [3], a chance-constrained planner with online replanning that bounds failure probability to a user-set risk; failure stayed within target in all tests, missions ~50–60% shorter than TERRA (failure up to 85%).
- Developed a precedence-aware planner [4] ($O(mn^3)$) that jointly plans routes and timing under energy and ordering constraints: 100 points, 10 teams in 0.34 s, with ~10x shorter makespan than a decoupled baseline.
- Built the air–ground exploration framework [2] (dual-layer PRM, coupled orienteering): 95% exploration 23–35% faster than adapted TERRA in simulation; hardware (Crazyflie + TurtleBot4) explored >95% of unknown space in 4 tours.
- Build and maintain the group's UAV-UGV hardware testbed (two Crazyflie UAVs, two TurtleBot4 UGVs, OptiTrack, ROS2) used in [1] and [2], and the ROS2 wind-disturbance simulation used in [3].

ME 5250: Robot Mechanics and Control, Northeastern University — *Boston, MA*
TEACHING ASSISTANT — *Fall 2026*

- Hold office hours and grade homework, midterms, and projects.

Miniature Robotics Lab (MRL), Bilkent University — *Ankara, Turkey*
UNDERGRADUATE RESEARCHER — *Feb. 2022 – July 2023*

- Design, fabrication, and locomotion analysis of Miniature Soft Quadruped (SQuad) robots.

ASELSAN — *Ankara, Turkey*
CANDIDATE ENGINEER — *Dec. 2022 – June 2023*

- Designed and optimized Kalman filters for motor control in SATCOM terminal and ground center systems.

Skills & Service

Programming	Python, MATLAB, Julia, C++
Robotics	ROS2, ROS, Gazebo, RViz, Simulink, Docker, Git, Linux; Crazyflie, TurtleBot4, OptiTrack, E-puck2
Planning & Control	Motion Planning, State Estimation, Multi-Agent Coordination, Task Allocation
Optimization & Theory	MIP (Gurobi), Vehicle Routing & Scheduling, Combinatorial Optimization, Complexity Analysis
Service	Reviewer: IEEE ICRA and American Control Conference (ACC), 2026